

Yu-Sheng Shih

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CURRENT POSITION

Sep. 2026–Present Research Assistant
Institute of Information Science, Academia Sinica Taipei, Taiwan
» *Supervisor:* Prof. [Meng-Tsung Tsai](#)
» Research focus: streaming graph algorithms and communication complexity

EDUCATION

Jan. 2025–Aug. 2026 M.S. in Computer Science and Information Engineering
National Taiwan University Taipei, Taiwan
» *Advisor:* Prof. [Shang-En Huang](#)

Aug. 2019–Jun. 2024 B.S. in Computer Science and Information Engineering
National Taiwan University Taipei, Taiwan
» Cumulative GPA: 4.05/4.3
» Double major in Mathematics

RESEARCH INTERESTS

- » Graph theory and graph algorithms
- » Streaming and sublinear algorithms
- » Communication complexity
- » Extremal and probabilistic combinatorics
- » Combinatorial optimization

RESEARCH PROJECTS

- » **Space Complexity Dichotomies for Subgraph Finding Problems in the Streaming Model**
» Yu-Sheng Shih, Meng-Tsung Tsai, Yen-Chu Tsai, and Ying-Sian Wu
» Established space-complexity dichotomies for subgraph and induced-subgraph finding in undirected and oriented graph streams.
» [PREPRINT](#) · [arXiv:2602.08002](#) (2026)
- » **SDP-Based Reductions for Coloring 3-Colorable Graphs via Short Odd-Cycle Elimination**
» [MASTER'S THESIS](#) · Completed Aug. 2026; related research ongoing
» *Advisor:* Prof. [Shang-En Huang](#)
» Developed SDP-based reductions that eliminate triangles and 5-cycles while retaining a large induced subgraph, reducing the coloring problem to instances of larger odd girth.

OTHER COAUTHORED WORK

- » Saintan Wu, Sen-Peng Eu, Kuo-Han Ku, and **Yu-Sheng Shih**. *Combinatorial proofs of Petrie Pieri rule and Plethystic Pieri rule*. *Annals of Combinatorics*, accepted for publication, 2026.
- » Tony Feng et al. (24 authors, including **Yu-Sheng Shih**). *Semi-Autonomous Mathematical Discovery with Gemini: A Case Study on the Erdős Problems*. arXiv:2601.22401, 2026.

AWARDS

- » **International Mathematical Olympiad**: Silver Medal (2019); Bronze Medal (2018)
- » **ICPC Taipei Regional**: Silver Prize (2021 and 2022)
- » **Asian Pacific Mathematics Olympiad**: Bronze Medal (2018)

INTERNSHIPS

Summer 2025	Research Intern, Streaming Algorithms Institute of Information Science, Academia Sinica	Taipei, Taiwan
Winter 2023	Quantitative Research Intern WorldQuant Taiwan	Taipei, Taiwan

TEACHING AND ACADEMIC SERVICE

Feb. 2026–Jun. 2026	Teaching Assistant — Graph Theory with Applications National Taiwan University	Taipei, Taiwan
Sep. 2025–Present	Founder and Organizer, NTU TCS Study Group National Taiwan University » <i>Cofounders</i> : Peng-Ting Lin, Ben Cheung, Shang-En Huang, and Po-Yu Chou	Taipei, Taiwan
Sep. 2021–Jan. 2022	Teaching Assistant — Algorithm Design and Analysis National Taiwan University	Taipei, Taiwan

SELECTED ACADEMIC ACTIVITIES

Jul. 2026	Participant, IMS Summer School National University of Singapore » <i>Innovations and Challenges in Extremal Combinatorics</i>	Singapore
Apr. 2026	Taiwan Team Observer A, European Girls' Mathematical Olympiad EGMO	Bordeaux, France
Jul. 2025	Participant, PCMI Graduate Summer School IAS/PCMI » Program: <i>Extremal and Probabilistic Combinatorics</i>	Park City, Utah, USA

LANGUAGES AND TECHNICAL SKILLS

Languages	Mandarin Chinese (native); English (TOEFL iBT 101, 2024); Japanese (JLPT N2, 2026)
Technical	C++, Python, $\text{\LaTeX}$, Linux